EXPERIMENT 7: CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

LEARNING OBJECTIVES:

- Understand the concept of **K-Means clustering**.
- Load and explore the **customer dataset** using Pandas.
- Select and standardize **Annual Income and Spending Score**.
- Find the optimal number of clusters using the **Elbow Method**.
- Apply **K-Means clustering** and assign customers to clusters.
- Visualize and analyze the **clusters and centroids**.

METHODS USED / PYTHON LIBRARIES INVOLVED:

Library/method	Purpose	Reason for Use
Pandas	Data loading, cleaning and manipulation	Efficiently handles tabular datasets
NumPy	Numerical operations	Useful for numerical calculations
Elbow Method	Optimal cluster selection	Finds the suitable number of clusters
StandardScaler	Data standardization	Scales features to a common range
K-Means	Customer clustering	Groups customers based on similarity

PROGRAM CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
df = pd.read_csv("Mall_Customers.csv"
```

```
print(df.head())
print(df.shape)
print(df.isnull().sum())
print(df.describe())

X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

wcss = []
for k in range(1, 11):
    model = KMeans(n_clusters=k, random_state=42, n_init=10)
    model.fit(X_scaled)
    wcss.append(model.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("K")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

model = KMeans(n_clusters=5, random_state=42, n_init=10)
df['Cluster'] = model.fit_predict(X_scaled)

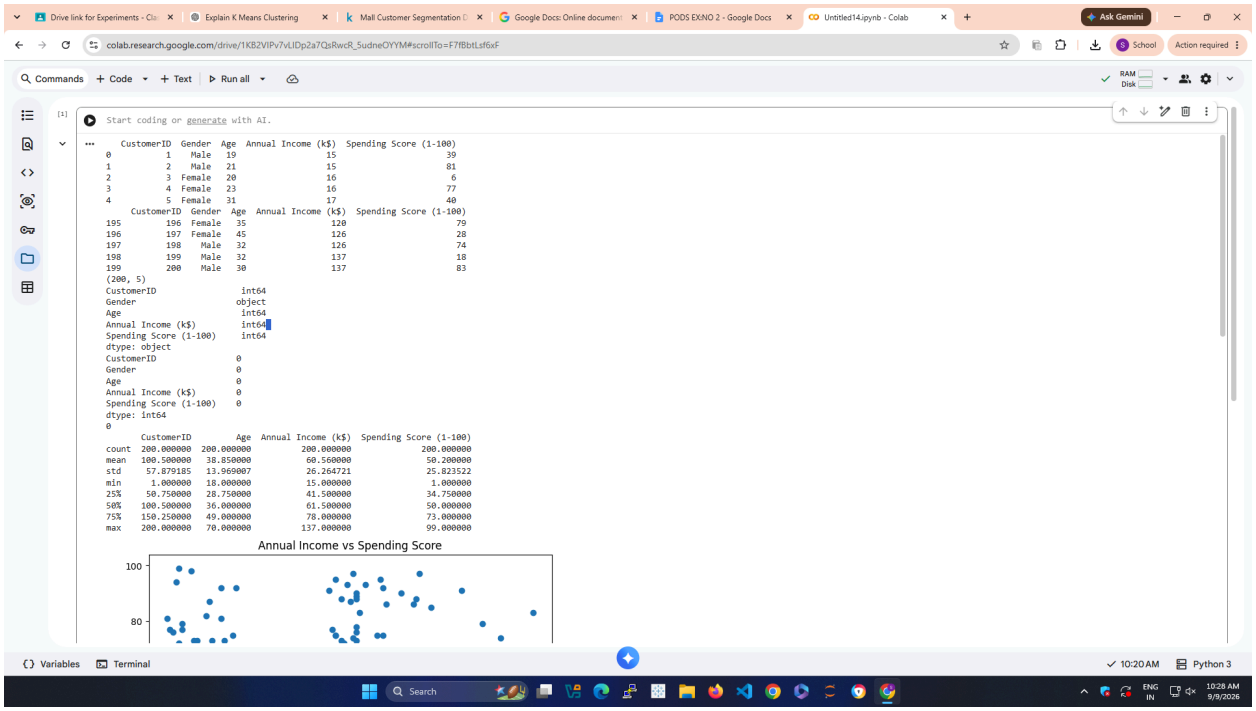
centers = scaler.inverse_transform(model.cluster_centers_)

plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=df['Cluster'])
plt.scatter(centers[:, 0], centers[:, 1], marker='X', s=200)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Segmentation")
plt.show()

print(df.groupby('Cluster').agg({
    'Annual Income (k$)': 'mean',
    'Spending Score (1-100)': 'mean'
}))

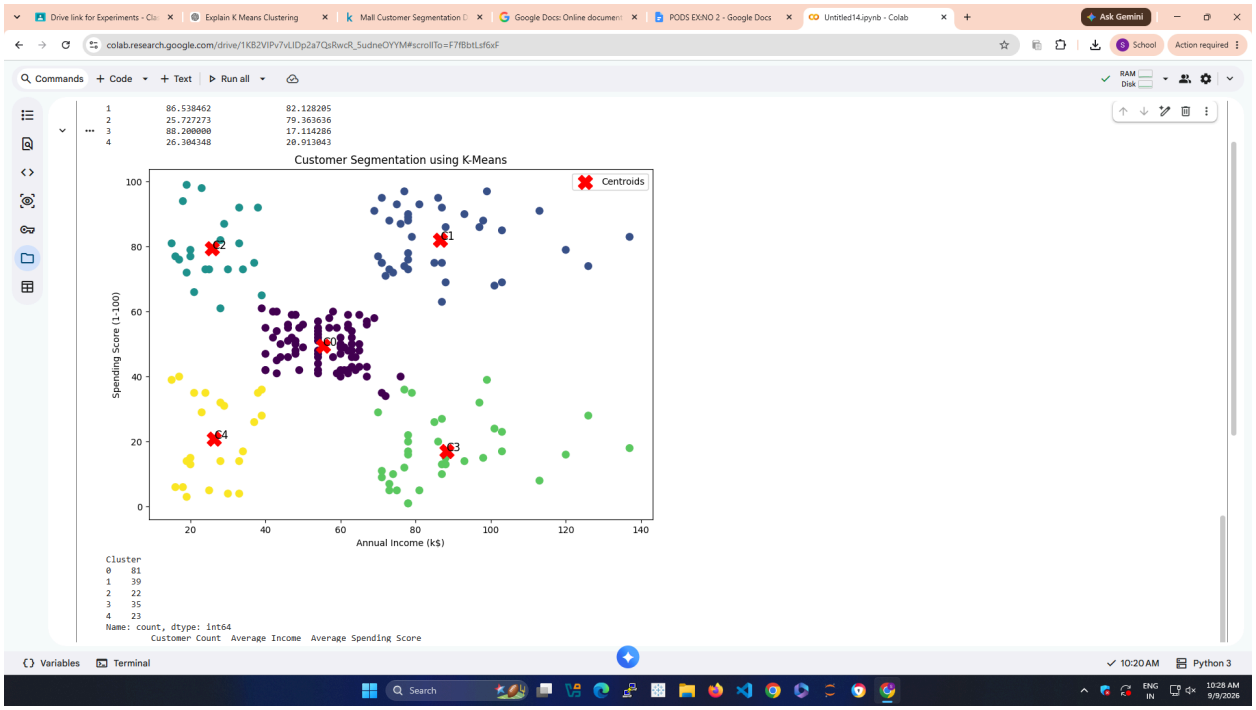
df.to_csv("Mall_Customers_Clustered.csv", index=False)
```

OUTPUT:



```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
     CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
195         196  Female   35                 120                      79
196         197  Female   45                 126                      28
197         198    Male   32                 126                      74
198         199    Male   32                 137                      18
199         200    Male   30                 137                      83
(200, 5)
CustomerID                int64
Gender                   object
Age                       int64
Annual Income (k$)        int64
Spending Score (1-100)    int64
dtype: object
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
0
       CustomerID         Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000          200.000000              200.000000
mean   100.500000   38.850000           60.560000               50.200000
std     57.879185   13.969007           26.264721               25.823522
min      1.000000   18.000000           15.000000                1.000000
25%     50.750000   28.750000           41.500000               34.750000
50%    100.500000   36.000000           61.500000               50.000000
75%    150.250000   49.000000           78.000000               73.000000
max    200.000000   70.000000          137.000000               99.000000
```



LEARNING OUTCOMES:

- Understand the concept of **K-Means clustering**.
- Perform **customer segmentation** using Annual Income and Spending Score.
- Apply **data standardization** using StandardScaler.
- Determine the **optimal number of clusters** using the Elbow Method.
- Visualize **clusters and centroids** using Matplotlib.
- Analyze the **characteristics of each customer cluster**.